UNIVERSITÀ DEGLI STUDI DI SALERNO

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE ED ELETTRICA E MATEMATICA APPLICATA

CORSO DI LAUREA MAGISTRALE IN INGEGNERIA INFORMATICA (LM-32)

TESI DI LAUREA

AUTOMATIC DETECTION OF FIGHT EVENTS IN VIDEO SEQUENCES

Relatori:

Ch.mo Prof. Mario Vento

Ch.mo Prof. Antonio Greco

Candidata:

Mariapia Lombardi

Matricola 0622701957

Anno Accademico 2024 – 2025

"To my family, who always supported me during this journey…"

Table of Content

1.Introduction 5
1.1 Problem Definition 5
1.2 Relevance to Computer Engineering 6
2. State-of-the-Art 11
2.1 Introduction to VAD 11
2.1.1 Supervised VAD 11
2.1.2 Weakly-Supervised VAD 12
2.1.3 Semi-Supervised / Self-Supervised VAD 14
2.1.4 Unsupervised VAD 15
2.2 Weakly Supervised Video Anomaly Detection 17
2.3 PI-VAD 20
2.4 Identification of potential improvements over the state-of-the-art 24
3. Original Contribution to the Problem Solution 27
3.1 Definition of the proposed methodology 27
3.2 Methodological innovations 28
3.3 Design of the Proposed System 29
3.4 Technological and/or practical innovations 31
3.5 Tools, technologies, and models used in the development 35
3.5.1 Vision Transformers 35
3.5.2 V-JEPA2 39
3.5.3 XD-Violence Dataset 43
3.5.4 Hardware and Software Resources 45
4. Experimental Validation and Application Aspects 48
4.1 Description of the Result Evaluation Metrics 48
4.1.1 Area Under the Curve (AUC) 49
4.1.2 Abnormal Area Under the Curve (AUC_A) 52
4.1.3 Average Precision (AP) 52
4.1.4 Abnormal Average Precision (AP_A) 53

4.1.5 Precision, Recall and F1-score 54
4.2 Experimental Protocol Definition 56
4.3 Data and Case Study Description for the Experiments 58
4.4 Presentation and Analysis of the Results 60
4.5 Comparison with State-of-the-Art 62
5. Conclusions and future works 68
Acknowledgments 72
List of Figures 73
List of Tables 74
References 75

1.Introduction

1.1 Problem Definition

Video Anomaly Detection (VAD) refers to the automatic identification of anomalous events within video sequences, where anomalies are defined as activities that deviate from behaviors considered normal in the reference context. In literature, such events are often also referred to as *abnormalities*, *outliers*, or *novelties*. Common examples of anomalies include violent behaviors, intrusions, thefts, falls, the presence of suspicious objects, or movements that are inconsistent with the scene. [1]

VAD can be regarded as a form of **spatiotemporal anomaly detection**, since an anomaly may manifest itself:

- **spatially**, as the presence of an unexpected object in a scene
- **temporally**, as a sudden change in motion patterns
- **spatiotemporally**, when anomalous activity arises from the combination of appearance and dynamics.

A frequently adopted formal definition describes an anomaly as

"The manifestation of atypical visual or motion characteristics, or the occurrence of typical patterns that, however, take place within a spatiotemporal context that does not conform to established norms."

For example, an anomaly may correspond to a traffic accident (an anomalous visual event), but also to riding a bicycle on a pedestrian sidewalk (a normal behavior occurring in an inappropriate context).

The nature of anomalies makes VAD a particularly challenging task: they are typically rare, difficult to define precisely and often characterized by

high variability. In addition, anomalies are highly context-dependent, as the same action or behavior may be considered normal or abnormal depending on the specific scene, environment, or application domain. Moreover, video data are high-dimensional signals that require the simultaneous modeling of both spatial and temporal information. [2]

This thesis focuses on a specific subclass of Video Anomaly Detection, namely the automatic detection of violent behaviors (*fight detection*). The task consists in analyzing video sequences and determining whether they contain episodes of physical violence between individuals. This type of event is characterized by irregular dynamics, abrupt movements, and close interactions among multiple subjects, which can make anomaly identification particularly challenging. The objective is therefore to design a system capable of reliably recognizing such episodes, despite the typical variability of real-world videos, such as non-uniform lighting conditions, heterogeneous viewpoints, limited recording quality, and the high variability of human behavior.

1.2 Relevance to Computer Engineering

The automatic detection of anomalies has become an imperative in many modern systems, where it is required to identify rare, unusual, or potentially dangerous events quickly and reliably. Video Anomaly Detection (VAD) has emerged as a key enabling technology across a wide range of application domains. These include intelligent video surveillance, smart city security, industrial automation, behavioral monitoring, healthcare assistance, autonomous driving, road surveillance, and environmental disaster detection and response.

In these contexts, VAD systems aim to continuously analyze video streams in order to identify deviations from normal patterns of activity that may indicate critical situations. For instance, in video surveillance scenarios, anomaly detection can automatically highlight violent behaviors, crimes, or accidents that require immediate attention. Similarly, in healthcare and assisted living environments, VAD techniques can be employed to monitor patients or elderly individuals, enabling the timely detection of falls, sudden distress, or other medical emergencies.

More broadly, the relevance of VAD in such domains stems from its ability to detect subtle or rare events that may signal security threats, medical emergencies, or catastrophic situations.

This growing demand has accelerated the development and adoption of advanced techniques based on Artificial Intelligence (AI), and in particular Deep Learning. Intelligent systems now play a central role in video processing, scene understanding, and the detection of anomalous behaviors, thanks to their ability to extract complex representations and model spatiotemporal dynamics that are difficult to capture with traditional methods. The integration of VAD and AI therefore represents a key aspect of contemporary digital transformation, enabling levels of automation and scalability that were previously unimaginable.

In the domain of video surveillance, anomaly detection constitutes a fundamental component of modern security systems, as it allows the automatic identification of suspicious or critical behaviors - such as assaults, thefts, accidents, or emergency situations - without relying exclusively on human monitoring. The continuous observation of video

streams is in fact a cognitively demanding task, prone to errors and inherently difficult to scale in real-world scenarios characterized by an ever-increasing number of cameras. Automated VAD systems can alleviate this burden by highlighting potentially relevant events, improving response times and reducing the risk of missed detections.

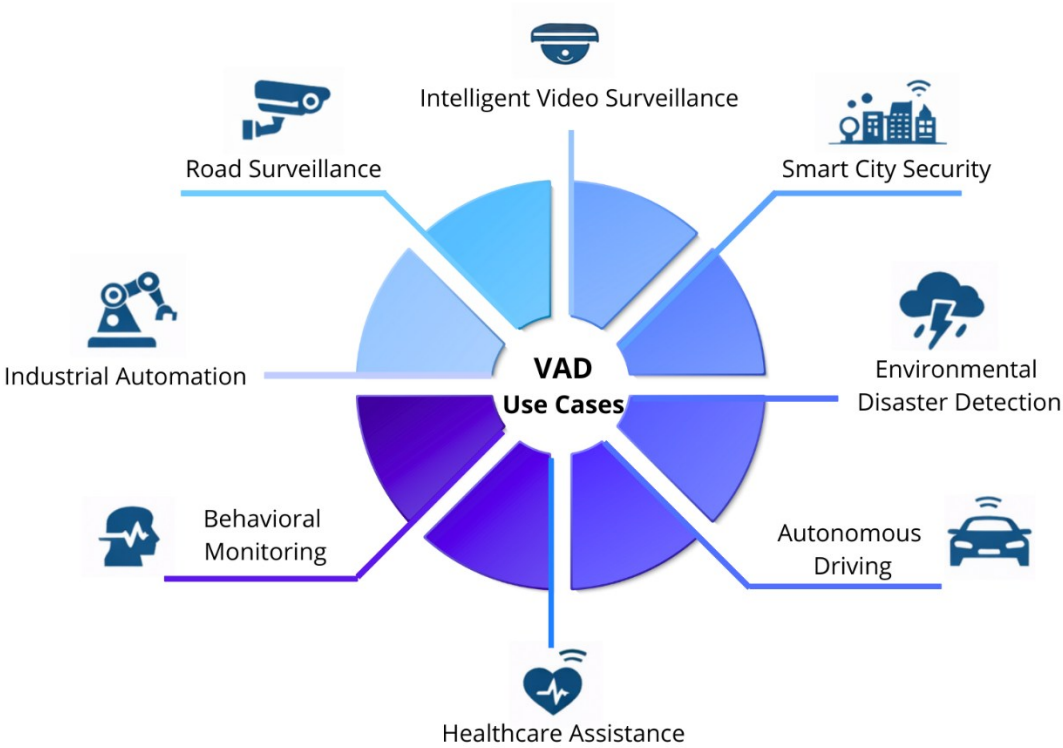

Figure 1. Video Anomaly Detection Use Cases

Despite its relevance, Video Anomaly Detection presents significant computational and engineering challenges. Anomalies are rare events that are difficult to define precisely, which complicates both their modeling and the training of reliable detection systems. Moreover, video data require the joint processing of spatial and temporal information: compared to static images, videos introduce additional complexity related to motion,

scene variability, and the dynamic evolution of events. Factors such as the low resolution typical of surveillance cameras, the scarcity of precise annotations, and the heterogeneity of operational environments further increase the difficulty of the task.

These challenges position VAD as a key area of contemporary computer engineering. Its objective is to develop high-performance deep learning models, video processing systems optimized for real-time operation, scalable infrastructures capable of monitoring hundreds of streams simultaneously, and mechanisms that enable continuous adaptation to changing environments. [3]

The remainder of this thesis is organized as follows:

- Chapter 2 reviews the state-of-the-art in Video Anomaly Detection (VAD), providing an overview of the main categories of VAD approaches based on different supervision levels, including supervised, weakly supervised, semi/self-supervised, and unsupervised methods. Particular emphasis is placed on weakly supervised VAD, which represents the most widely adopted paradigm, with a discussion of recent advances in this area. The chapter concludes by identifying the main limitations of existing methods and the potential improvements that motivate the proposed approach.
- Chapter 3 presents the automatic fight recognition system proposed in this thesis, describing the adopted methodology, the main innovations introduced, and the tools and technologies used for its implementation.

- Chapter 4 describes the experimental setup and evaluation protocol, and reports the results obtained from the conducted experiments, providing a quantitative and qualitative analysis of the performance of the proposed system.
- Chapter 5 concludes the thesis by summarizing the main contributions of this work, discussing its limitations, and outlining possible directions for future research and development.

2. State-of-the-Art

2.1 Introduction to VAD

Given the intrinsic complexities of VAD discussed in Chapter 1, it becomes necessary to balance the models' ability to correctly detect anomalies (accuracy) with their applicability at large scale without excessive annotation or computational costs (practical feasibility). In response to this challenge, the research community has developed approaches with varying levels of supervision: from fully supervised methods, which require detailed annotations, to weakly supervised or unsupervised solutions, capable of leveraging more partial information or data-driven statistical distributions [2].

The following sections provide an overview of the main levels of supervision adopted in VAD.

2.1.1 Supervised VAD

In **Supervised VAD methods**, the model is trained on fully annotated datasets in which both normal and anomalous behaviors are explicitly labeled. Within this setting, the VAD problem is generally formulated as a classification task, where the model is trained to distinguish between normal and anomalous events.

This approach is particularly effective in contexts where anomalies are well defined and annotated data are available in sufficient quantities, such as in certain healthcare or industrial applications. Thanks to the availability of precise labels, supervised methods can learn highly discriminative features, often achieving high performance in the detection of known anomalies.

However, the applicability of fully supervised methods is severely limited in real-world scenarios. Anomalies are rare events and are inherently imbalanced with respect to normal data, making the construction of representative datasets challenging. In addition, the strong context dependency of anomalous events and the difficulty of unambiguously defining what constitutes an anomaly significantly increase the complexity of the task. The same behavior may be normal in one setting but anomalous in another. For example, a punch may represent an act of violence in a public space, yet be perfectly acceptable in a boxing gym.

Therefore, accurate video annotation - often required at fine-grained temporal levels - is a costly and time-consuming process, frequently requiring the involvement of domain experts.

Finally, supervised models typically suffer from limited generalization capabilities in open-set scenarios, resulting in poor performance when detecting novel or previously unseen anomalies. For these reasons, despite their strong performance in controlled settings, supervised methods exhibit limited scalability in real-world Video Anomaly Detection applications.

2.1.2 Weakly-Supervised VAD

Weakly supervised methods have emerged as a response to the limitations of fully supervised approaches, particularly the scarcity of accurate annotations and the intrinsic ambiguity in the definition of anomalies. Obtaining precise labels - such as frame-level or even pixel-level annotations - is an extremely costly process and often requires specialized expertise. In many contexts, including clinical monitoring and

behavioral analysis, identifying the exact onset and offset of anomalous behavior requires expert intervention, making the creation of fully annotated datasets impractical.

As discussed previously, anomalies are highly context-dependent, which complicates the annotation process even further.

To address these limitations, weakly supervised approaches rely on weak labels, typically provided at the video level. The most widely adopted paradigm in this context is **Multiple Instance Learning (MIL)**, introduced by Sultani et al. [4]. In this framework, each video is divided into segments (or instances) and is labeled as anomalous if it contains at least one anomalous segment, without specifying its temporal location (see Figure below).

While one of the main advantages of these methods is the absence of a requirement for detailed annotations, a key limitation lies in the risk of focusing solely on the most prominent anomaly, potentially overlooking more subtle patterns.

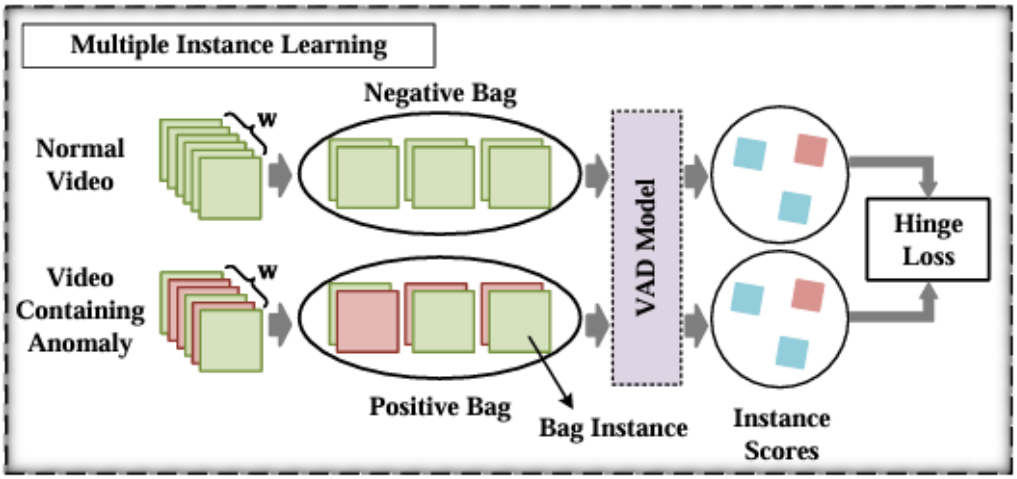

Figure 2. VAD formulated as a weakly supervised problem using MIL.

2.1.3 Semi-Supervised / Self-Supervised VAD

Semi-supervised methods offer a good compromise between accuracy and annotation requirements, positioning themselves between the supervised and unsupervised paradigms. In this approach, during training, only videos containing normal behaviors are used, with the goal of learning a robust representation of "normal behavior". Anomalies are then detected as significant deviations from the learned patterns.

In the past, these approaches were often classified as unsupervised; however, more recent studies have clarified that they incorporate an implicit form of supervision: the training dataset is divided into normal sequences (used to learn the distribution of normal behavior) and anomalous sequences (not used during training). This minimal yet meaningful distinction justifies their classification as semi-supervised rather than unsupervised.

Moreover, many semi-supervised methods also fall under the broader paradigm of **self-supervised learning**, in which supervisory signals are not derived from manual annotations but from auxiliary tasks generated directly from the normal data. In this way, during training, the model autonomously learns the regular structure of video sequences, while at inference, sequences that produce high reconstruction errors or exhibit low likelihood are considered anomalous.

Depending on the learning objective, semi- and self-supervised methods can be classified into four main categories:

1. **Reconstruction-based**: The model (often an autoencoder) reconstructs normal sequences. Anomalies are identified through high reconstruction errors.

2. **Prediction-based**: The model learns to predict the future behavior of normal videos. Anomalous events are detected when the prediction significantly deviates from the observed data.

3. **Jigsaw Puzzle**: An auxiliary task is formulated by shuffling video segments (spatial, temporal, or both). The model's ability to correctly solve the puzzle is used as a measure of normality; sequences that are difficult to reconstruct are considered anomalous.

4. **Distribution Estimation**: The model estimates the distribution of normal data; instances with low likelihood relative to the learned distribution are flagged as anomalies.

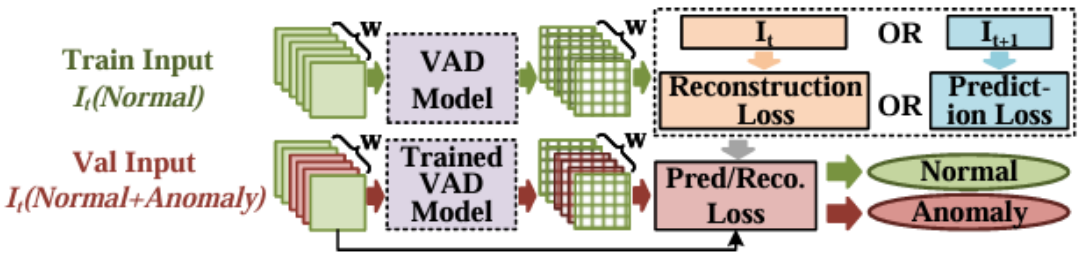

Figure 3. Self/semi-supervised VAD achieved through reconstruction or prediction.

2.1.4 Unsupervised VAD

In **fully unsupervised methods**, no labels or prior assumptions about the nature of the data (normal or anomalous) are available. This category represents the "pure" form of anomaly detection and remains relatively underexplored, as many works that self-identify as unsupervised are in fact based on semi- or self-supervised approaches, trained exclusively on normal data and therefore implicitly containing label information.

The key principle of unsupervised methods is the so-called “normality advantage”: it is assumed that most of the data represents normal behavior, while anomalies are rare and irregular events. By leveraging this statistical imbalance, models learn the global characteristics and structure of normal data, while significant deviations are interpreted as anomalies.

Among the most used paradigms are:

- **Clustering**: Normal data tend to form dense clusters in feature space, while isolated points or points in low-density regions are considered anomalies.
- **Pseudo-label Induction**: The model generates pseudo-labels by exploiting reconstructions or predictions of the data itself, iteratively refining the separation between normal and anomalous instances. Although conceptually similar to semi- and self-supervised methods, here the labels are self-generated without access to ground truth.

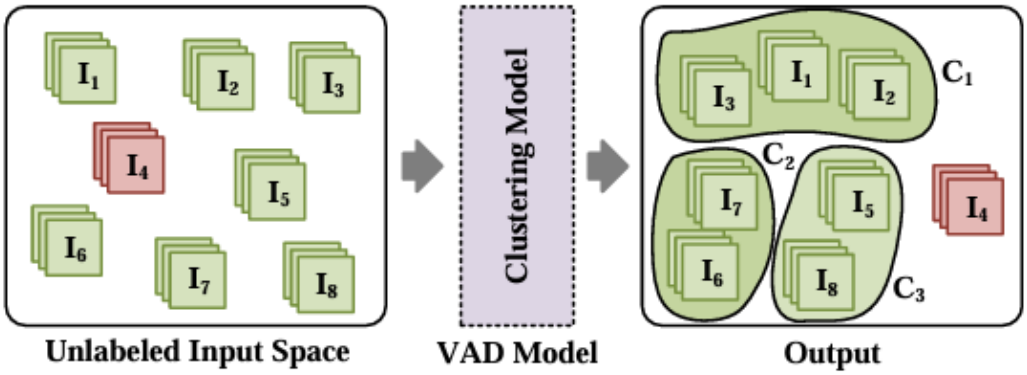

Figure 4. Unsupervised anomaly detection through clustering.

2.2 Weakly Supervised Video Anomaly Detection

According to the classification of Video Anomaly Detection methods, it becomes evident that **weakly supervised approaches** represent an effective compromise between accuracy (the ability to correctly identify anomalous events) and practical feasibility (the cost and complexity of annotation). In many real-world scenarios, it is indeed possible to obtain labels at the video-sequence level - such as an indication of the presence or absence of an anomalous event - while precise temporal annotation of anomalies at the frame level are prohibitive in terms of cost, time, and required expertise.

This setting also reflects the structure of many publicly available datasets for VAD, in which the training set is typically annotated at the video level, while more detailed frame-level annotations are provided exclusively for the test set for evaluation purposes. This design choice significantly reduces the burden of manual annotation, making large-scale model training feasible.

Weakly supervised methods therefore allow the exploitation of partial yet realistic information, maintaining greater flexibility than fully supervised approaches and better discriminative capability than methods based solely on modeling normal behavior.

However, while this paradigm avoids the costly frame-by-frame annotation process, it also introduces significant challenges in terms of accuracy, temporal localization, and interpretability of the model's decisions.

For these reasons, this section is devoted to the analysis of the main **Weakly-Supervised Video Anomaly Detection (WSVAD)** approaches proposed in the literature.

WSVAD methods have proven particularly effective in detecting large-scale and visually salient anomalies, such as explosions, traffic accidents, or sudden crowd movements. However, these approaches face greater challenges in identifying more complex, human-centered anomalies - such as thefts, assaults, abuses, or fights - where interactions among multiple subjects, subtle movements, and intricate temporal dynamics play a crucial role.

One of the main causes of these limitations lies in the fact that many existing solutions rely on a single source of information, typically RGB visual data, which may not be sufficient to fully capture the semantic and temporal complexity of such scenarios. As a result, accurate localization and reliable interpretation of this type of anomaly remain open challenges. [5]

From a methodological perspective, the foundational work by Sultani et al. [4] introduced a deep learning–based **Multiple Instance Learning (MIL)** framework for VAD. Within this paradigm, each video is divided into segments (instances) and labeled as anomalous if it contains at least one anomalous instance, without providing information about its temporal location. During training, the model therefore learns to autonomously identify which parts of the video are responsible for the anomaly, reducing the cost of supervision and better handling semantic ambiguity.

Building upon this pioneering contribution, numerous subsequent studies have proposed extensions and improvements to the MIL framework, with

the goal of enhancing temporal localization accuracy, system robustness, and overall performance under weak supervision. Among these, Tian et al. [6] introduced a feature-magnitude-based learning function to emphasize anomalous instances, while Chen et al. [7] proposed a feature amplification mechanism combined with an amplitude contrast loss to strengthen the discriminative power of the learned representations. Lv et al. [8] instead presented an Unbiased Multiple Instance Learning (UMIL) framework, aimed at reducing training-induced biases and achieving more balanced anomaly classifiers.

Despite the improvements introduced, these "one-stage" approaches tend to focus primarily on the most discriminative segments, often overlooking more ambiguous or subtle anomalies. To address this limitation, more recent works have shifted attention toward "two-stage" strategies based on self-training and pseudo-labeling [9] [10], with the aim of progressively refining anomaly scores. In this context, Li et al. [10] proposed a multi-sequence learning technique that enables the iterative refinement of anomaly estimates through the progressive selection and reduction of the considered sequences.

However, these methods also predominantly rely on visual information from a single modality, limiting the model's ability to capture the semantic complexity of anomalous events. To overcome this restriction, recent studies have begun to explore "cross-modal" approaches, integrating information from multiple modalities to improve the quality of both feature representations and pseudo-labels. In most cases, such methods exploit textual anomaly categories, which nevertheless fail to fully represent the semantic richness and contextual complexity of real-world events.

The **multi-modal video representation learning** paradigm aims to construct more informative representations by combining multiple modalities, such as RGB, depth, audio, text, or pose. This approach mainly relies on two strategies:

- **contrastive loss**, used for instance in models such as CLIP [11] to learn a shared embedding space across different modalities
- **knowledge distillation**, which enables the transfer of knowledge from a more informative modality - such as text - to a visual modality like RGB.

Some approaches, such as ViFiCLIP [12] and CoCLR [13], combine both techniques to improve cross-modal alignment and enhance learning efficiency.

Despite the potential offered by these solutions, the use of multimodality in the context of Weakly-Supervised VAD remains limited.

First, there exists a significant semantic gap between different modalities, which capture information at varying levels of abstraction and make their effective integration challenging. Second, many multimodal models exhibit high computational complexity, assuming the simultaneous availability of all modalities even during inference. This requirement leads to substantial computational costs, often incompatible with real-time application scenarios typical of video surveillance.

2.3 PI-VAD

To overcome the limitations of single-modality weakly supervised methods and, at the same time, the high computational complexity typical

of traditional multimodal models, Majhi et al. proposed the **PI-VAD**, or **π-VAD (Polymodal Induced Video Anomaly Detection)** framework [5], which represents an innovative contribution in the context of Weakly-Supervised VAD.

The core idea behind π-VAD is to enrich RGB video representations by inducing information from five complementary modalities:

- **pose**, to capture fine-grained human body movements
- **depth**, to provide three-dimensional representations of the scene and objects
- **panoptic masks**, to encode contextual and semantic segmentation information of scene elements
- **optical flow**, to model global motion
- **linguistic signals** derived from vision-language models, capable of providing high-level semantic descriptions of events.

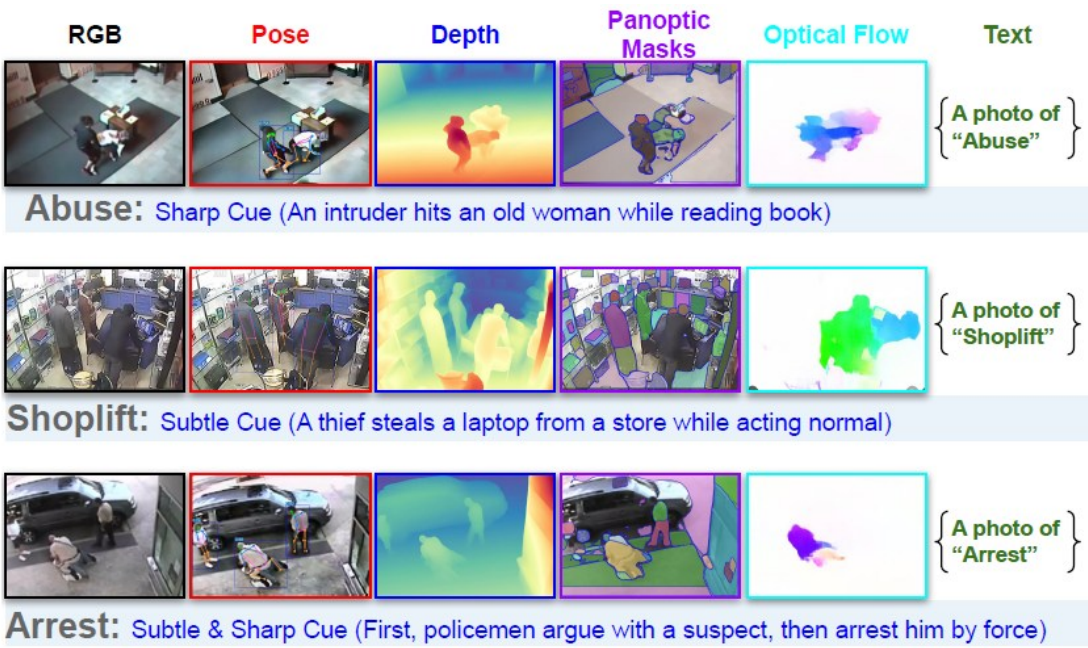

Figure 5. Illustration of abnormal frames and respective multi-modal saliencies in complex real-world scenes.

Each modality can be interpreted as an axis of a multimodal polygon, designed to provide complementary informative cues to the RGB signal.

The framework is based on a **teacher-student architecture**, in which both models share the same architectural structure but perform different roles.

The teacher model is pre-trained on the VAD task in a weakly supervised setting and is subsequently kept frozen. The student model, randomly initialized, is instead the only model updated during training and is responsible for the final learning process.

The key contribution of π-VAD is the **Poly-Modal Inductor**, which consists of two fundamental modules: the **Pseudo Modality Generation (PMG)** module and the **Cross-Modal Induction (CMI)** module.

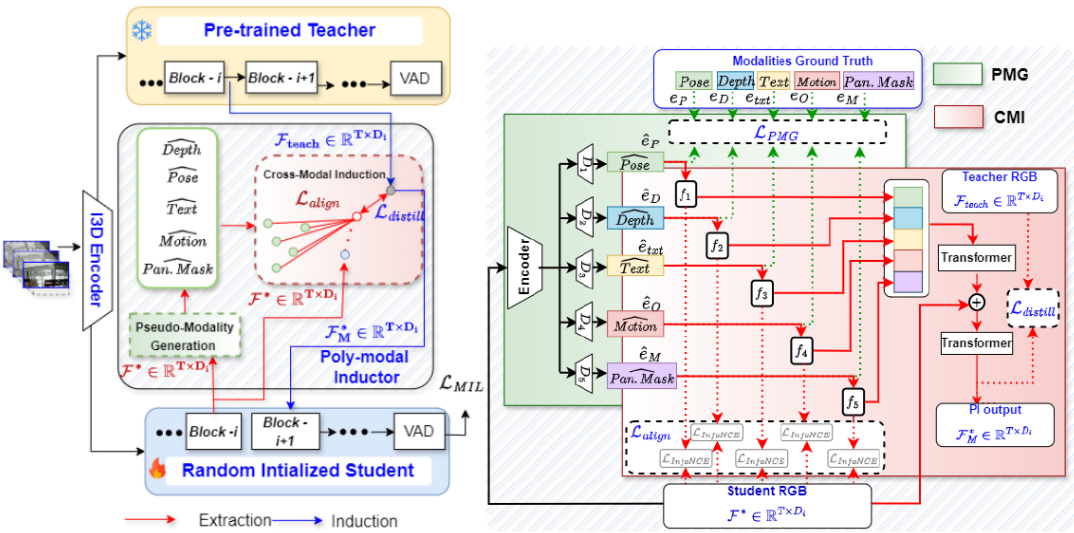

Figure 6. Poly-modal Induced VAD framework.

As shown in Figure 6, the PMG module aims to synthetically derive embeddings for pose ($\widehat{e_P} \in \mathbb{R}^{T \times d_P}$), depth ($\widehat{e_D} \in \mathbb{R}^{T \times d_D}$), panoptic masks ($\widehat{e_M} \in \mathbb{R}^{T \times d_M}$), optical flow ($\widehat{e_O} \in \mathbb{R}^{T \times d_O}$), and text ($\widehat{e_{txt}} \in \mathbb{R}^{T \times d_{txt}}$)

directly from the student's intermediate RGB feature representation $F^* \in \mathbb{R}^{T \times D_i}$.

This mechanism overcomes the limitations of relying solely on the RGB signal without introducing explicit multimodal backbones, thereby preserving high computational efficiency.

The CMI module, instead, is responsible for semantically aligning the generated pseudo-modalities $\hat{e}_j$ with the RGB features F^* within a shared embedding space anchored to the primary visual modality. This alignment is achieved through a dual strategy: on the one hand, a contrastive loss encourages coherent associations between each pseudo-modality and the RGB representation; on the other hand, a teacher-guided knowledge distillation process ensures that the induced multimodal information is effectively relevant to the anomaly detection task.

The Poly-Modal Inductor is also flexible and can be integrated into different blocks of the teacher–student architecture; however, it is deliberately placed in the early and final blocks to capture both low-level multimodal features and high-level semantic representations.

A significant strength of π-VAD lies in its computational efficiency: the pseudo-modalities are exploited exclusively during the training phase, while during inference the model operates solely on the RGB stream. This design choice makes the framework particularly suitable for real-time scenarios and easily scalable, while still benefiting from the rich information typically associated with multimodal models.

The π-VAD method has been evaluated on several major public datasets for Video Anomaly Detection, including **UCF-Crime** [4] and **XD-Violence** [14]. In both cases, multiple types of multimodal information are

considered, such as pose, depth, semantic cues, textual content, and motion. In the case of the **XD-Violence** dataset, which also includes audio tracks, this information is treated as an additional modality and aligned with the others through the Polymodal Induction mechanism.

Additional experiments were conducted on the **MSAD** dataset [15], a more recent collection of real-world anomalous videos characterized by more heterogeneous scenarios compared to UCF-Crime, in order to evaluate the model's generalization capability across different operational contexts.

Experimental results demonstrate that π-VAD achieves state-of-the-art performance across all three benchmarks considered, while maintaining high inference efficiency, as it does not require the explicit use of multimodal backbones [5]. Considering these characteristics, π-VAD represents one of the first systematic attempts to effectively integrate multimodality within the context of Weakly-Supervised Video Anomaly Detection, constituting an important reference point in the state of the art.

2.4 Identification of potential improvements over the state-of-the-art

Prior work on Video Anomaly Detection demonstrates that methods belonging to the WSVAD paradigm have achieved significant results in terms of accuracy, particularly in the detection of visually salient and large-scale anomalies. At the same time, several limitations have emerged, highlighting areas for further improvement.

A first critical aspect concerns **the nature of the detection task**. Most WSVAD methods address the problem as a binary classification between

normal and anomalous videos, where “anomalous” is interpreted in a general sense, merely indicating deviation from normality, without providing insight into the specific type or category of anomaly. While this approach is suitable for quantitative performance evaluation, it is less effective in real-world application scenarios, where it is often necessary to understand the type of detected anomalous event to support operational decision-making, reduce false alarms, and prioritize interventions.

A second relevant aspect concerns **the video representations employed**. Many state-of-the-art approaches rely primarily on RGB features, which may be insufficient to capture the complexity of events characterized by rapid movements, multiple interactions, and contextual variations. Although recent works have explored the integration of multimodal information to enrich the semantic description of scenes, such solutions are often limited by high computational complexity or by the need for dedicated backbones for each modality, making their adoption in real-time or large-scale scenarios challenging.

Another key aspect relates to **the balance between accuracy and practical feasibility** offered by weakly supervised methods compared to other supervision paradigms. Fully supervised approaches, while potentially capable of achieving high performance on known anomalies, are difficult to scale due to the high cost of fine-grained temporal annotation; as a result, only a few public datasets provide frame-level labels for training. Conversely, approaches based solely on modeling normal behavior may suffer from limited discriminative capacity when faced with complex anomalies. In this context, weakly supervised methods represent an effective intermediate solution, but they introduce

new challenges related to precise temporal localization of anomalies and the interpretation of weak supervisory signals.

Considering these limitations, there is a clear need to explore solutions capable of:

1. **Going beyond simple binary anomaly detection,** which only identifies the presence of an anomaly without specifying its type, and focusing instead on specific categories of anomalous events that are relevant from an application perspective
2. **Exploiting more expressive video representations**, capable of effectively modeling temporal dynamics and interactions between subjects
3. **Maintaining a balance between performance, robustness, and computational cost**, ensuring applicability in real-time and large-scale scenarios.

This work is positioned within this context, focusing on the automatic detection of fight episodes in video sequences. This choice allows for the evaluation of models not only in distinguishing between normal and anomalous behaviors, but also in capturing complex behavioral patterns characterized by strong interactions between individuals, irregular movements, and a high degree of context dependency.

3. Original Contribution to the Problem Solution

3.1 Definition of the proposed methodology

This work addresses the problem of Video Anomaly Detection, with a specific focus on the automatic recognition of physical violence between individuals (fight detection). The goal is to design a system capable of analyzing video sequences and determining the presence of violent behaviors, while addressing the main challenges typical of real-world video surveillance scenarios, such as scene variability, the complexity of human interactions, and the limited availability of precise temporal annotations.

The proposed methodology formulates fight detection as a class-specific anomaly recognition problem, explicitly targeting physical violence between individuals rather than performing generic discrimination between normal and anomalous events. This formulation enables a shift from generic anomaly scoring toward a more semantic and task-oriented understanding of events, thereby increasing the practical relevance of the system in security and automated monitoring applications.

From a learning perspective, the proposed methodology adopts a fully supervised paradigm, leveraging temporally annotated video data to enable precise clip-level classification of violent events.

The proposed approach leverages a video model pre-trained on large-scale collections, which is subsequently adapted to the fight detection task through supervised fine-tuning, allowing the system to exploit robust spatiotemporal representations and transfer this knowledge to recognize violent behaviors.

The overall design of the methodology ensures compatibility with standard evaluation protocols, facilitating direct comparison with existing methods and a clearer interpretation of the experimental results presented in the following sections.

3.2 Methodological innovations

This section highlights the main methodological innovations aimed at improving the recognition of violent episodes in real-world video sequences.

As discussed in Section 3.1, a key feature of the proposed system is its **focus on a specific anomaly class** that is **fight detection**, which allows the model to learn features relevant to violent behaviors rather than performing a generic normal/anomalous distinction.

Another methodological contribution concerns the adoption of a **fully supervised learning paradigm**. The fully supervised setting adopted in this work was made possible by a careful frame-level annotation of the training set used in this study, which will be introduced and described in detail later in the thesis. This allows the model to be directly optimized on temporally coherent clips, improving the accuracy of fight recognition and supporting precise temporal localization of violent events.

A further innovation relates to the **end-to-end adaptation of the model to the fight detection task**. While many existing solutions rely on pre-computed video features fed into separate classifiers, in the proposed approach the model receives raw video inputs and performs both feature extraction and prediction within a single pipeline. This enables representations to be learned and fine-tuned directly for the specific task,

improving the system's ability to capture complex spatiotemporal interactions and motion dynamics characteristic of violent behaviors.

3.3 Design of the Proposed System

This section describes the overall architecture of the proposed fight detection system, which is organized as a sequence of modular components, from video acquisition to the generation of clip-level predictions.

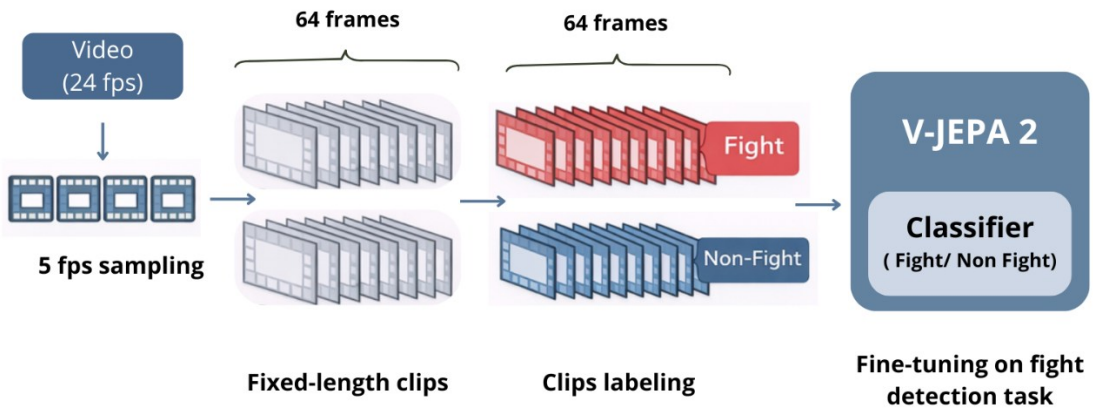

Figure 7. Overview of proposed model.

In the first stage, input videos were preprocessed to match the requirements of the pipeline. Initially, frames were sampled at 5 fps, a rate chosen to preserve sufficient temporal information for capturing violent events. The sampled frames were then grouped into fixed-length clips of 64 consecutive frames, corresponding to approximately 13 seconds, considering that the original videos have a frame rate of 24 fps. This clip length was selected to match the input size expected by the model responsible for human fight detection, ensuring proper processing and compatibility with its architecture.

Prior to model processing, the clips were normalized and resized to ensure uniform dimensions within training batches. The final resizing to 384×384 pixels was handled internally by the model.

As a next step, labels were assigned to each clip based on the frame-level annotations created for the dataset used in this work, namely XD-Violence [14]. Specifically:

"a clip was labeled as containing a fight if at least 15% of its frames depicted violent behavior; otherwise, it was considered normal."

This threshold was chosen to ensure that a clip was labeled as containing a fight only if the event was present and clearly visible. Since the XD-Violence dataset contains multiple types of anomalous events in addition to fights, labeling an entire clip as containing a fight when only a small portion of its frames depicts violent behavior could have provided misleading supervision to the model. In such cases, the model would have been forced to learn fight patterns from clips that largely did not contain this event and that might instead represent normal behavior or other types of anomalies. By adopting this criterion, the labeling process maintained both accuracy and reliability, providing the model with more consistent supervisory signals.

Then, the processed clips were then fed to the model responsible for fight detection, namely V-JEPA2 [16], which had been pre-trained on the Something-Something-v2 (SSv2) dataset [17]. This pre-training provided rich spatiotemporal representations that were sensitive to the dynamics of human actions, forming a strong foundation for subsequent fine-tuning on the fight detection task.

Within the proposed pipeline, the model was adapted to the fight detection task through a fully supervised fine-tuning. Specifically, the classification head and the final self-attention layer were updated, while the remaining layers were left frozen. The original classification head, trained on 174 action classes in SSv2, was replaced with a binary head producing predictions “fight vs non-fight”. This end-to-end adaptation enabled the model to perform both feature extraction and classification directly from raw video inputs, improving its ability to capture complex spatiotemporal interactions and motion dynamics while maintaining computational efficiency.

Details regarding the training procedure, including learning rate, batch size, early stopping, and number of epochs used, are provided in Chapter 4, where the experimental protocol is described.

During inference, the system processes videos on a clip-by-clip basis, producing predictions that can be used to temporally localize violent events. This design makes the pipeline suitable for monitoring long video streams and supports applications in real-time scenarios.

3.4 Technological and/or practical innovations

A key technological and practical innovation introduced in this work is the **manual frame-level annotation of the training set of the XD-Violence dataset**, which was a fundamental prerequisite for adopting a fully supervised learning paradigm in the V-JEPA2 model.

In its original form, the dataset provides video-level labels for the training set and frame-level labels only for the test set. In particular, the dataset included an annotation file associated with the test set, in which each line

corresponded to a single anomalous video. For each video, the file reported the temporal intervals (in terms of start and end frames) during which anomalous events occurred. The figure below shows an excerpt of the original annotation file:

```
v=7xQr5wYwNPg__#1_label_B4-0-0 0 403 600 1008
v=-fOWSLV6Esw__#1_label_B4-0-0 20 772
v=b0MUjeKAGZw__#1_label_B4-0-0 0 2790
Bad.Boys.1995__#01-11-55_01-12-40_label_G-B2-B6 157 180 185 244 250 360 582 810
Bad.Boys.1995__#01-33-51_01-34-37_label_B2-0-0 57 670 728 800
Bad.Boys.II.2003__#00-06-42_00-10-00_label_B2-G-0 2610 2670 2760 3060 3167 4310
v=DD3jfKr8e-k__#00-00-00_00-01-40_label_B4-0-0 0 2400
v=2jwO15SMyuo__#1_label_B4-0-0 0 2374
v=jYENhkzdpO8__#00-00-00_00-01-31_label_B6-0-0 245 335 1187 1252 1836 2000
v=1djrJ0wxlYo__#1_label_B4-0-0 0 758 880 1533
v=7zEFHKKBA0g__#1_label_B4-0-0 1192 1600 2031 2690
```

Figure 8. Original annotation structure used in the test set of XD-Violence dataset.

In the original annotation file, anomalous events are represented using a set of predefined labels.

Specifically, **B1** corresponds to fights, **B2** to shootings, **B4** to riots/protests, **B5** to abuse, **B6** to car accidents and **G** to explosions. Videos without any anomalies are labeled as **A**, although in the annotation file only videos containing anomalous events are listed.

For example, the line:

Bad.Boys.II.2003__#00-06-42_00-10-00_label_B2-G-0 2610 2670 2760 3060 3167 4310

represents a video titled *"Bad.Boys.II.2003__#00-06-42_00-10-00"*, where the anomalous events present are encoded in *_label_B2-G-0*. Each video contains up to three types of anomalies, which are indicated after the *_label* prefix and separated by "-"; if a video has fewer than three types of anomalies, zeros are used to fill the remaining positions.

As can be observed from the above figure, in the original test set annotations, association between the specific type of anomaly and the frame interval wherein it was contained was not explicitly reported. This limitation affects only videos containing multiple types of anomalies. Furthermore, there was no one-to-one correspondence between the anomalies listed in the *_label* field and the frame intervals: a single anomaly could occur multiple times within a video, and the anomalies were not listed in the order in which they appeared. As a result, for videos with multiple anomalous events, the original annotation format did not allow determining which specific anomaly occurred within a given frame interval, making frame-level analysis ambiguous.

To address this limitation, prior to annotating the training set, the test set annotations were refined. Specifically, each annotated frame interval was augmented by explicitly indicating, in parentheses, the type of anomaly occurring within that interval. This refinement was performed by visually inspecting the corresponding video and assigning the appropriate anomaly class to each interval, resulting in a more informative and semantically rich annotation format.

The figure below illustrates an example of the updated annotations file:

```
v=7xQr5wYwNPg__#1_label_B4-0-0 0 403 600 984
v=-fOWSLV6Esw__#1_label_B4-0-0 20 772
v=b0MUjeKAGZw__#1_label_B4-0-0 0 2795
Bad.Boys.1995__#01-11-55_01-12-40_label_G-B2-B6 157 180 (B2) 185 244 (G) 250 273 (B2) 288 336 (G)
Bad.Boys.1995__#01-33-51_01-34-37_label_B2-0-0 96 670 728 800
Bad.Boys.II.2003__#00-06-42_00-10-00_label_B2-G-0 2610 2670 (B2) 2760 3024 (B2) 3144 3240 (B2) 32
v=DD3jfKr8e-k__#00-00-00_00-01-40_label_B4-0-0 0 2400
v=2jwO15SMyuo__#1_label_B4-0-0 0 2376
v=jYENhkzdpO8__#00-00-00_00-01-31_label_B6-0-0 245 335 1187 1252 1836 2000
v=1djrJ0wxlYo__#1_label_B4-0-0 0 758 880 1536
v=7zEFHKKBA0g__#1_label_B4-0-0 1192 1600 2031 3600
```

Figure 9. Updated test set annotations file of XD-Violence dataset

This refinement process was applied only to videos containing multiple types of anomalies; in videos with a single type of anomaly, all frame intervals clearly corresponded to that unique anomaly, so no additional information was required.

The same annotation structure was then adopted for the training set to ensure full consistency between training and test data. Each anomalous video of the training set was visually inspected to identify the exact frame intervals in which anomalous events occurred. For each interval, defined by a pair of start and end frames, the corresponding type of anomaly was explicitly recorded in parentheses, mirroring the refined annotation format used for the test set.

This procedure produced precise and temporally accurate labeling of the training videos, allowing each frame to be unambiguously associated with the presence or absence of specific anomalous behaviors. Such detailed annotations enabled the construction of reliably labeled temporal clips and made it possible to train the model in a fully supervised manner, with direct supervision at clip-level.

As a result of this annotation process, the system was able to exploit precise temporal supervision, better capturing rapid movements, complex interactions, and contextual variations characteristic of fight events. This led to improved accuracy in fight recognition as well as more reliable temporal localization of violent episodes.

Overall, this frame-level annotation effort represents a significant technological and practical contribution. This approach made it possible to adopt training strategies and model adaptations that would have been impossible using only the original video-level annotations.

3.5 Tools, technologies, and models used in the development

For the development of the proposed system, a set of software tools and deep learning models was employed, selected for their capability to process video data, learn complex spatiotemporal representations and ensure computational efficiency.

3.5.1 Vision Transformers

Since their introduction in 2017 by Vaswani et al. [18], Transformers have established themselves as the state of the art for Natural Language Processing (NLP) tasks.

At the core of this architecture lies the self-attention mechanism, which allows the model to dynamically weigh the relative importance of different elements within a sequence when computing their representations. By relying entirely on this mechanism, Transformers demonstrated a superior ability to model long-range dependencies in sequential data, overcoming key limitations of traditional sequence models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs). Unlike recurrent architectures, Transformers avoid the sequential processing bottleneck, enable efficient parallelization during training and are better suited to preserve information over long temporal spans, quickly becoming the reference architecture for sequence-to-sequence problems such as machine translation.

In 2021, Dosovitskiy et al [19] successfully adapted the Transformer architecture to Computer Vision tasks, introducing the **Vision Transformer (ViT).**

The ViT reformulates an image as a sequence of fixed-size patches, which are linearly embedded and processed in a manner analogous to word tokens in NLP.

While the original Transformer architecture consists of both an encoder and a decoder, the Vision Transformer employs only the encoder stack. The output of the encoder is then passed to a task-specific neural network head, which produces the final prediction. The attention mechanism remains the core component, enabling the model to capture global relationships across the entire image.

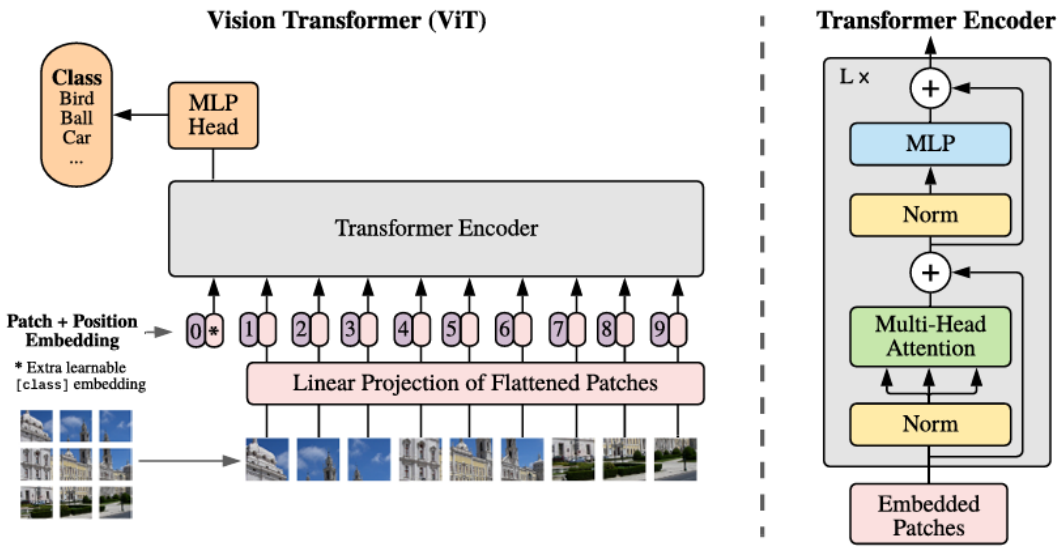

Figure 10. Overview of the Vision Transformer (ViT)

The first step of the ViT is to create tokens from the input image. Transformers operate on a sequence of tokens; in NLP, this is commonly a sentence of words.

In case of ViT, an image is converted to tokens such that each token represents a local area – or patch – of the image.

More specifically, to handle 2D images, the image $x \in \mathcal{R}^{H \times W \times C}$ is reshaped into a sequence of flattened 2D patches $x_p \in \mathcal{R}^{N \times (P^2 \cdot C)}$, where (H, W) is the resolution of the original image, C is the number of channels, (P, P) is the resolution of each image patch, and $N = \frac{HW}{P^2}$ is the resulting number of patches, which also serves as the effective input sequence length for the Transformer.

After extracting tokens from an image, it is common to use a linear projection to change the length of the tokens. This is implemented as a learnable linear layer. The new length of the tokens is referred to as the *latent dimension*, *channel dimension*, or the *token length.*

After the projection, the tokens are no longer visually identifiable as a patch from the original image.

The output of this projection is referred to as the *patch embeddings*.

After the tokens creation, the next two steps of the ViT, before the encoding blocks, are referred to as *token processing*.

During this phase, the first step is to prepend a blank token, called the *Prediction Token,* to the image tokens. This token will be used at the output of the encoding blocks to make a prediction. It is initialized as a zero vector, allowing it to progressively aggregate information from the other image tokens through the self-attention mechanism.

Since the Transformer architecture does not include any inherent mechanism to model the order of input elements, positional information must be explicitly provided.

In Vision Transformers, this is achieved through *positional embeddings*, which encode the spatial arrangement of image patches and allow the model to distinguish between different spatial configurations.

Positional embeddings are represented as learnable vectors with the same dimensionality as the patch embeddings and are added element-wise to the input sequence. This additive formulation preserves the dimensionality of the representations while enriching them with spatial information.

During pretraining, and optionally during fine-tuning, positional embeddings are learned jointly with the rest of the model parameters, progressively capturing the underlying spatial structure of the image.

After that, tokens are ready to proceed to the encoding blocks.

The encoding blocks represent the core computational component of the Vision Transformer, where the model learns interactions between image tokens.
The number of encoding blocks is a hyperparameter defined by the model configuration. A diagram of the encoding block is shown in Figure 8.

After passing through the encoding blocks, the last thing the model must do is make a prediction.

Throughout the encoding block(s), the prediction token has become nonzero and gained information about the input image. Only this prediction token is used to make a final prediction.

Finally, the prediction token is passed through the *head* to make a prediction. The *head*, usually some variety of neural network, varies based on the model. For example, in [19], an MLP with one hidden layer during

pretraining and a single linear layer during fine tuning is used. In [20], a single linear layer is used as a head.

3.5.2 V-JEPA2

The backbone of the proposed pipeline is the V-JEPA2 model [16], a world model developed by Meta that can understand videos, predicting future events, and planning actions, demonstrating unprecedented zero-shot capabilities.

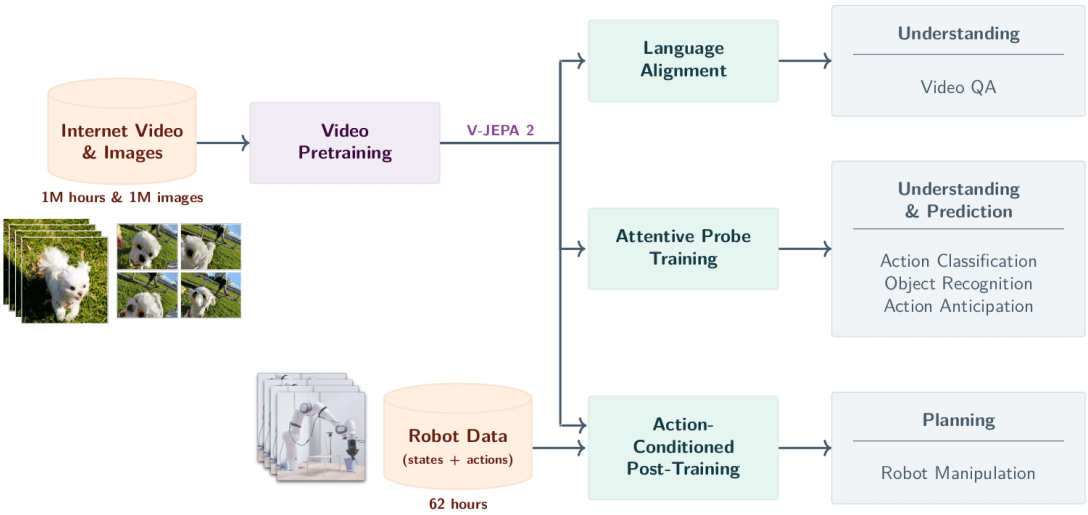

Figure 11. V-JEPA2 architecture overview

V-JEPA2 is based on a Vision Transformer (ViT) architecture, which was discussed in the previous section, and consists of two main components: an encoder and a predictor.

The encoder acts as the perception engine and is responsible for transforming raw video clips into abstract and meaningful feature vectors. To process video data, each input is divided into three-dimensional patches, referred to as *tubelets* (each consisting by default of 2 frames of

size 16×16 pixels), which serve as the fundamental tokens for the model. Using 3D Rotary Position Embeddings (3D-RoPE), the encoder is able to dynamically encode the spatial and temporal position of each *tubelet*, improving training stability for large-scale models. The encoder outputs a sequence of high-dimensional vectors, each representing a *tubelet* and containing detailed contextual information about the video.

The predictor, by contrast, serves as the learning engine and is implemented as a smaller ViT. Its role is to learn world dynamics by predicting the masked portions of the video based on the visible context provided by the encoder.

V-JEPA2 stands out for its strong capabilities in understanding scene dynamics, predicting future states, and planning actions in novel scenarios, relying solely on video data or limited amounts of interactive data.

Another key strength of the model lies in its multimodal flexibility: the video encoder can be aligned with language models for tasks such as video question answering, achieving state-of-the-art performance on benchmarks that require physical understanding and temporal reasoning.

For the fight detection task, the V-JEPA2 model employed in this work is the version pre-trained on the Something-Something-v2 (SSv2) dataset [17]. Developed by a professional team at Qualcomm, this dataset constitutes a fundamental resource for action recognition–oriented video understanding models, owing to its design philosophy and scalability.

Figure 12. Samples from the Something-Something dataset

SSv2 comprises over 220,000 videos divided into 174 distinct action classes, characterized by detailed and temporally dependent human-object interactions. An example image of the classes used in the dataset is shown below:

"Putting something on a surface"	"Turning something upside down"
"Moving something up"	"Tearing something into two pieces"
"Covering something with something"	"Pushing something from left to right"
"Pushing something from left to right"	"Squeezing something"
"Pushing something from right to left"	"Throwing something"
"Uncovering something"	"Putting something next to something"
"Taking one of many similar things on the table"	"Poking something so lightly that it doesn't or almost doesn't move"

Figure 13. Sample Classes

Unlike other datasets with generic labels such as “sports” or “cooking,” SSv2 classes describe highly specific actions, such as “Folding something,” “Placing something on a surface,” “Pretending to grab something,” or “Flipping something.”

A distinctive aspect of SSv2 is its strong emphasis on interactions rather than the mere presence of objects. The dataset trains models to differentiate between gestures that are visually similar but semantically different, such as “picking up a cup,” “placing a cup,” “sliding a cup,” and “pretending to drink from a cup.” This forces the model to capture subtle temporal cues and motion dynamics that define each action, rather than merely identifying co-occurring objects.

Furthermore, SSv2 places particular emphasis on the temporal structure of actions: the beginning, middle, and end of each video clip are crucial for correct classification. For instance, the action “Opening something” appears significantly different at the start compared to the end. This makes SSv2 an ideal training ground for models like V-JEPA2, which are designed to predict the temporal evolution of scenes. V-JEPA2’s predictive approach, aiming to anticipate future video segments based on past context, aligns perfectly with the requirements of a dataset so rich in temporal information.

Pre-training on SSv2 provides V-JEPA2 with a solid foundation of spatiotemporal representations, particularly suited to capturing movements and interactions in violent scenes.

3.5.3 XD-Violence Dataset

Regarding the dataset used for training and evaluating the fight detection system, XD-Violence [14] was selected - a large-scale, multi-scene dataset representative of real-world contexts.

The dataset consists of 4,754 unedited videos, totaling approximately 217 hours of content, accompanied by audio signals and weak video-level annotations. Specifically, it includes 2,405 violent videos and 2,349 non-violent videos, split into 3,954 videos for training and 800 videos for testing.

XD-Violence covers six categories of violent events (abuse, traffic accident, explosion, fight, riot, and shooting), in addition to normal scenes, and employs multi-label annotations that allow multiple types of violence to occur simultaneously within the same video. The videos are highly diverse, sourced from heterogeneous platforms such as movies, YouTube, cartoons, sports events, video games, news broadcasts, CCTV, wearable cameras, and dashcams, providing a wide range of scenarios and visual conditions.

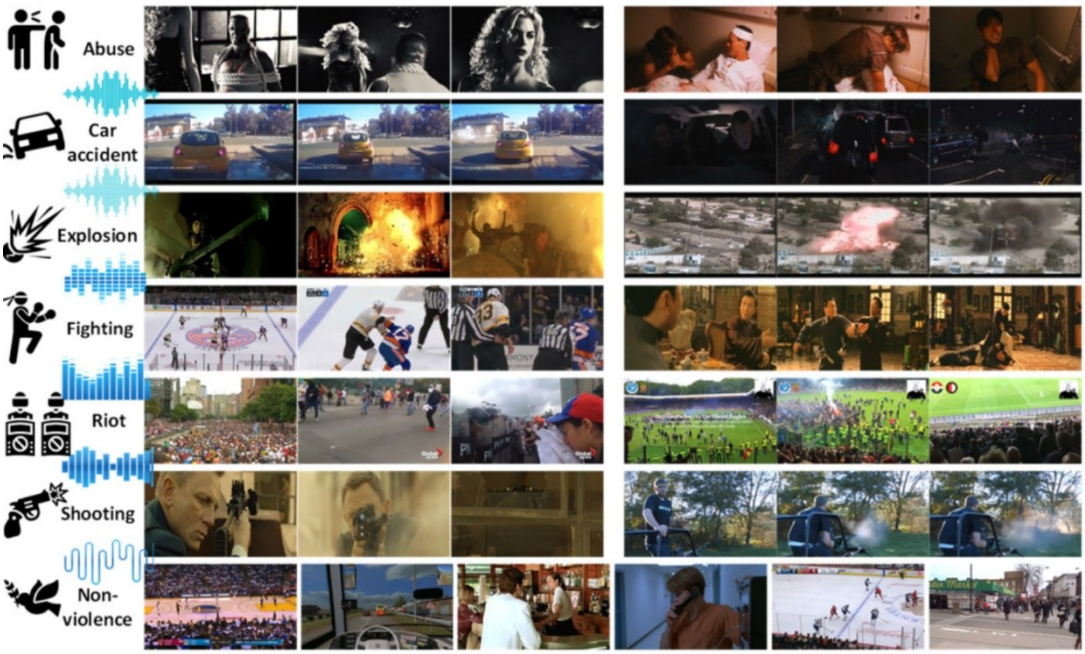

Figure 14. Sample videos from the XD-Violence dataset

The state-of-the-art annotations follow an official division: the training set provides video-level labels, indicating whether a video contains violent episodes, without specifying their temporal locations; the test set, on the other hand, provides frame-level annotations, which are useful for precisely identifying the time intervals in which violent events occur.

As already described in section 3.4, in this work the training set was further enriched through manual frame-level annotation to provide more accurate supervisory signals. This step was necessary to allow the division of original videos into fixed-length clips and to train the V-JEPA2 model with clip-level consistent labels, thereby improving the system's ability to recognize complex events such as fights.

3.5.4 Hardware and Software Resources

The entire system was implemented in Python, using the PyTorch framework for defining, training, and evaluating the V-JEPA2 model for the specified task. The *Hugging Face Transformers* library [21] was employed to utilize the model, providing pre-trained implementations of state-of-the-art models and dedicated tools for video sequence preprocessing. In particular, the library enabled loading the V-JEPA2 model in a pre-trained version and its adaptation to the specific task, handling the reconfiguration of the classification head for a binary task.

For video management and decoding, the *Decord* library [22] was used, a solution specifically designed for deep learning applications on video data. *Decord* is a high-performance library that allows efficient video reading, offering fast and low-latency access to individual frames. This feature is particularly relevant in neural network training contexts, where non-sequential access to frames is often required.

One of the main challenges in processing video data for deep learning is the need for random access to frames during training. Traditional video libraries are generally optimized for sequential playback, whereas machine learning pipelines frequently require irregular temporal sampling and extraction of frame subsequences. *Decord* addresses this issue directly, providing an infrastructure designed to efficiently support the access patterns typical of video training pipelines.

Another key advantage of *Decord* is its support for both CPU and GPU decoding, allowing, when available, the use of hardware acceleration to significantly reduce video reading and preprocessing times. This feature

contributes to improving the overall performance of the pipeline, especially for large-scale datasets and high-resolution videos.

In this work, the *VideoReader* class of the *Decord* library was employed, providing essential functionalities for accessing video content. It allows obtaining the total number of frames in a video and directly accessing frames at arbitrary positions via indexing. This functionality proved essential for the preprocessing phase, in which the original videos were sampled at 5 fps and then divided into fixed-length temporal clips (64 frames each), according to the input requirements of the V-JEPA2 model.

The extracted clips were subsequently stored in HDF5 format using the *h5py* library [23], adopting a hierarchical structure based on groups and attributes to compactly and consistently associate the frames, labels, and metadata of each clip.

During training, these clips were then decoded using the *PIL* library [24]. The HDF5 format is widely used in scientific computing and deep learning for managing large volumes of heterogeneous data, thanks to its ability to organize complex information into a single, efficient, and easily indexable structure.

Using HDF5 significantly reduced input/output (I/O) overhead during training, avoiding the need to repeatedly decode the original videos at each epoch. In this way, the preprocessed clips can be loaded directly into memory in a structured form, improving data access speed and making the training pipeline more stable and efficient.

Moreover, the ability to store frames in compressed format (JPEG) within HDF5 files allowed disk usage to be minimized while maintaining good

visual quality of the data, providing a favorable trade-off between storage space and reading times.

The experiments were conducted using a GPU provided by the DIEM Department at the University of Salerno, specifically a MIG (Multi-Instance GPU) partition of the SR675 server, equipped with 16 GB of GPU memory.

This hardware constraint directly influenced some design choices during the training phase. To this end, to simulate a larger effective batch size without increasing memory usage, a gradient accumulation strategy was employed, allowing gradients to be accumulated over multiple consecutive batches and updating the weights only after a predefined number of iterations.

At the same time, a mixed precision strategy was adopted during training, which reduces memory consumption and improves computational efficiency, positively impacting overall training time without degrading model performance.

Overall, using a shared GPU required careful optimization of the training pipeline, aiming to find a balance between model complexity and the available computational resources. This approach enabled effective model training and the conduction of reliable experiments despite the hardware limitations.

4. Experimental Validation and Application Aspects

4.1 Description of the Result Evaluation Metrics

This section describes the metrics used to evaluate the performance of the proposed fight detection system. These metrics are commonly employed in VAD literature and were chosen to ensure a robust and meaningful assessment in contexts characterized by highly imbalanced data and the presence of rare events.

The use of standardized metrics widely adopted in VAD domain allows for a direct and consistent comparison between the proposed system and state-of-the-art approaches, enabling an objective analysis of the system's strengths and potential limitations relative to existing solutions in literature.

Specifically, the metrics employed include the Area Under the Curve (AUC), the Average Precision (AP), and their variants Abnormal AUC (AUC_A) and Abnormal Average Precision (AP_A).

Additionally, classical binary classification metrics such as Precision, Recall, and F1-score were considered, providing a more detailed description of the model's behavior and highlighting the trade-off between the ability to correctly detect violent events and the false alarm rate.

Beyond accuracy metrics, measures focused on evaluating the computational efficiency of the system were also introduced, with the goal of assessing its feasibility in real-time application scenarios. Indicators such as inference latency - measured in milliseconds as the time required by the model to process an input clip and produce a prediction - as well as the number of frames and clips processed per second, were considered.

These metrics are essential in video surveillance and continuous monitoring contexts, where the timeliness of detection is as crucial as classification accuracy.

A detailed description of the metrics used is provided below.

4.1.1 Area Under the Curve (AUC)

The **Area Under the Curve (AUC)** [25] is a scalar measure of the overall performance of a binary classifier and is defined as the area under the Receiver Operating Characteristic (ROC) curve. The ROC curve represents the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) as the decision threshold applied to the model's predictions varies. In the ROC plot, the TPR is shown on the vertical (y) axis, while the FPR is shown on the horizontal (x) axis.

TPR, also known as *sensitivity*, is the ratio between the number of positive samples correctly identified by the model and the total number of available positive samples. FPR, also referred to as the *false alarm probability*, corresponds to the ratio between the number of negative samples incorrectly classified as positive and the total number of truly negative samples. Each operating point of the model corresponds to a point on the curve.

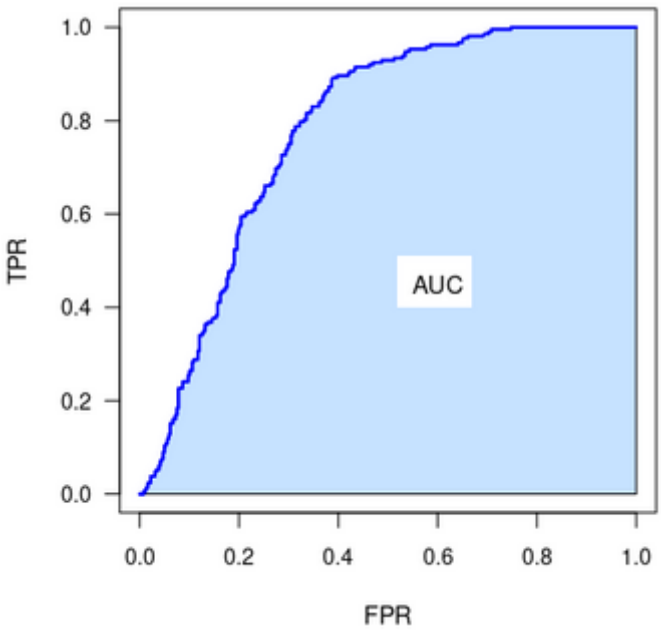

Figure 15. Area Under the ROC Curve.

An ideal classifier would reach the top-left corner of the plot (coordinates 0,1), corresponding to zero false negatives and zero false positives, thus achieving perfect classification. In contrast, a random classifier produces points along the diagonal from the bottom-left corner to the top-right corner, commonly referred to as the *line of no discrimination*.

The position of the points relative to the diagonal provides an intuitive indication of the classifier's quality: points above the diagonal indicate performance better than random guessing, while points below the diagonal correspond to worse-than-random performance.

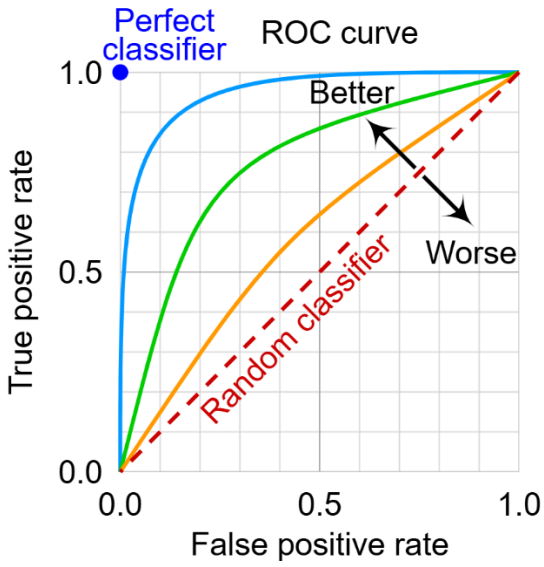

Figure 16. The ROC space for a "better" and "worse" classifier.

AUC values close to 1 suggest a greater ability to distinguish between classes, whereas values close to 0.5 correspond to performance comparable to random classification.

In the context of Video Anomaly Detection, the AUC measures the model's ability to correctly discriminate between normal video segments and segments containing anomalies, independently of the choice of a specific decision threshold.

This property makes AUC a widely adopted metric in VAD literature and suitable for comparing different approaches under a common evaluation protocol. However, in highly imbalanced scenarios - where anomalous events are extremely rare compared to normal behavior - AUC may offer a partially optimistic view of performance, as it gives equal importance to both classes and may not fully reflect the model's effectiveness in detecting rare anomalies.

4.1.2 Abnormal Area Under the Curve (AUC_A)

The **Abnormal Area Under the Curve (AUC_A)** is an extension of the AUC metric and is specifically designed to provide a more rigorous evaluation in Video Anomaly Detection scenarios. Unlike standard AUC, AUC_A is computed by considering only anomalous videos - i.e., those containing both normal and abnormal segments - while fully normal videos are excluded from the evaluation.

This choice allows the evaluation to focus on a more realistic and challenging setting, in which the model is required not merely to distinguish between normal and abnormal videos, but to correctly localize anomalies within complex video sequences characterized by the coexistence of lawful and unlawful behaviors.

In this way, AUC_A reduces the influence of "easy" cases, where the correct classification of entirely normal videos could artificially inflate performance, and instead provides a more informative measure of the system's actual ability to detect anomalies in application-relevant contexts.

4.1.3 Average Precision (AP)

Average Precision (AP) [26] summarizes the model's performance by jointly considering precision and recall and is defined as the area under the Precision - Recall (PR) curve. The PR curve describes how precision varies as a function of recall as the decision threshold applied to the model's predictions changes.

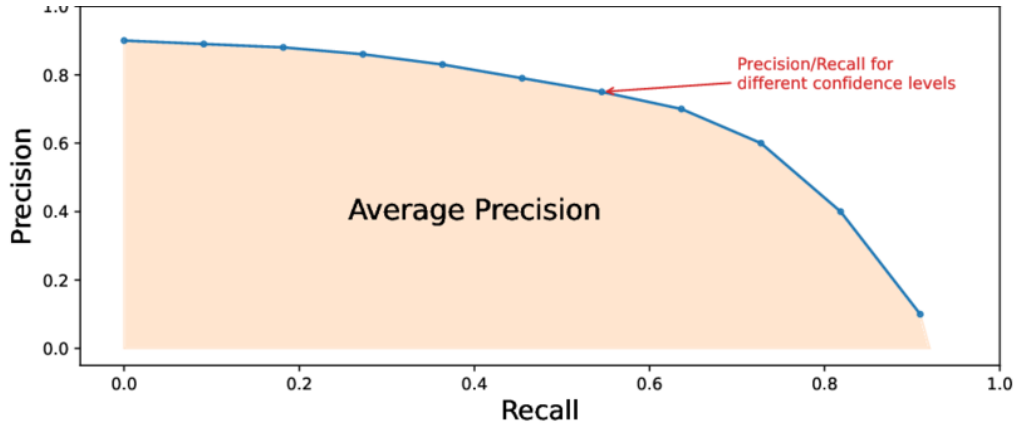

Figure 17. The Precision-Recall curve

This metric is particularly well suited to highly imbalanced settings, such as Video Anomaly Detection, where the anomalous class is significantly underrepresented compared to the normal one. In such scenarios, ROC-based metrics may be less informative, whereas AP provides a more informative and realistic evaluation of the system's performance, as it emphasizes correct detection of violent episodes rather than overall class separability.

Average Precision measures how effectively the system maximizes anomaly detection while keeping the number of false alarms low, thus serving as an indicator of the model's overall reliability in surveillance and security applications.

4.1.4 Abnormal Average Precision (AP_A)

Similarly to AUC_A, **Abnormal Average Precision (AP_A)** restricts the evaluation to anomalous videos only, excluding those that are entirely normal. For each video containing at least one anomalous event (in this case, a fight between individuals), Average Precision is computed by

considering both the normal and anomalous segments within the same video.

The final AP_A value is obtained by averaging the AP scores computed over individual anomalous videos. This approach enables performance evaluation in a more realistic setting, where the primary objective is not merely to determine whether a video contains violence, but to accurately localize anomalous events over time within complex sequences.

AP_A therefore represents a particularly suitable metric for comparing Video Anomaly Detection models, as it more faithfully reflects their ability to identify relevant events in scenarios where such events actually occur.

4.1.5 Precision, Recall and F1-score

To complete the analysis, the **Precision, Recall**, and **F1-score** [27] metrics were also considered.

Precision measures the ratio between the number of clips correctly classified as violent (i.e., True Positives) and the total number of clips predicted as violent by the model (given by the sum of True Positives and False Positives). It therefore provides an indication of the reliability of positive predictions and is particularly relevant in application scenarios where a high number of false positives could lead to unnecessary alarms.

$$Precision = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Positives\ (FP)}$$

Recall, instead, quantifies the system's ability to correctly identify clips containing violent episodes, by measuring the ratio between the number of violent clips that are correctly recognized (True Positives) and the total number of truly anomalous clips in the dataset (given by the sum of True Positives and False Negatives). A high Recall value indicates a system capable of reducing the number of missed violent events, which is a crucial aspect in video surveillance and security contexts.

$$Recall = \frac{True\ Positives\ (TP)}{True\ Positives\ (TP) + False\ Negatives\ (FN)}$$

The F1-score, defined as the harmonic mean of Precision and Recall, provides an overall and balanced measure of the classifier's performance. This metric is particularly informative in scenarios where it is necessary to strike a balance between reducing false positives and minimizing false negatives, enabling a concise assessment of the system's effectiveness in the presence of imbalanced data.

$$F1 = 2 * \frac{Precision * \ Recall}{Precision + Recall}$$

Overall, the joint use of global curve-based metrics, such as AUC, AUC_A, AP and AP_A, together with decision-threshold–dependent metrics, including Precision, Recall, and F1-score, enables a comprehensive evaluation of the system's performance. The former allowed for analyzing the model's ability to correctly rank video segments according to their degree of anomaly, independently of a specific threshold, while the latter provided more operational insights into the classifier's behavior under

realistic usage conditions, highlighting the trade-off between prediction reliability and the ability to detect violent events.

4.2 Experimental Protocol Definition

To evaluate the performance of the proposed fight detection system, an experimental protocol was defined to ensure both the reproducibility of the experiments and the correctness of the evaluation, in line with established practices in the context of Video Anomaly Detection (VAD).

Regarding the data used for the supervised fine-tuning of the V-JEPA2 model, the official split of the XD-Violence dataset was adopted, which divided the data into a training set and a test set. The training set was subsequently partitioned into two distinct subsets, used respectively for model training and its validation, with an 80%–20% split of the available data.

This partition was performed using a stratified split with respect to the fight class, in order to ensure that a representative number of violent events was present in the validation set as well. This choice was necessary given the strong rarity of violent episodes in the dataset, which could have resulted in a validation set devoid of anomalous samples, thereby compromising the ability to reliably monitor the training process.

Regarding the training setup, the supervised fine-tuning of the V-JEPA2 model was performed using the Adam optimizer with a differentiated learning rate strategy, in line with the partial unfreezing scheme adopted in this work.

As described in Section 3.3, only the classification head and the final self-attention layer of the V-JEPA2 model were updated during the fine-tuning, while all remaining layers were kept frozen.

A learning rate of 1×10^{-5} was assigned to the final self-attention layer. This choice was motivated by the need to slightly adapt the high-level spatiotemporal representations learned during pre-training to the fight detection task, while avoiding excessive modifications that could degrade the general representations learned by the backbone.

Conversely, a higher learning rate of 1×10^{-3} was used for the classification head, which was newly initialized and specifically designed for the binary classification task ("fight" vs "non-fight"). The higher learning rate allowed this component to learn discriminative decision boundaries more rapidly, without being constrained by the pre-trained weights.

Due to GPU memory constraints, discussed in Section 3.5, it was not possible to employ a large batch size during fine-tuning process. For this reason, gradient accumulation was adopted to simulate a larger effective batch size. Specifically, a physical batch size of 4 clips was used, with an accumulation step of 4, resulting in a virtual batch size of 16 clips. During training, a forward and backward pass was performed for each physical batch, while model parameters were updated only after accumulating gradients over 4 consecutive batches. This approach enabled more stable gradient updates while remaining within the available hardware limitations.

The model was trained for a maximum of 50 epochs. To reduce the risk of overfitting and to account for the scarcity of violent samples, an early

stopping strategy was employed based on the evaluation of AP_A metric, with a patience of 5 epochs. Training was terminated when no improvement in AP_A was observed for 5 consecutive epochs.

As already mentioned, the experimental evaluation followed standard practices in the Video Anomaly Detection literature, adopting widely used metrics such as AUC and AP, together with their abnormal variants (AUC_A and AP_A), in order to assess both detection performance and temporal localization of violent events.

In particular, the AP_A metric was used as a reference during the validation phase, as it focuses exclusively on positive samples and is more representative of the model's performance in the presence of rare events and highly imbalanced data.

As a complement of the analysis, Precision, Recall, and F1-score metrics were also considered, providing a more detailed evaluation of the classifier's behavior at the clip-level and of the trade-off between false positives and false negatives.

Finally, computational efficiency indicators were considered to assess the feasibility of the proposed system in real-time scenarios, including inference latency and the number of frames and clips processed per second.

4.3 Data and Case Study Description for the Experiments

While the XD-Violence dataset has been introduced in Section 3.5 from a technical perspective, this section discusses the reasons for its selection as an experimental case study for the proposed fight detection system.

XD-Violence represents one of the main reference benchmarks in the context of Video Anomaly Detection in real-world scenarios, due to its high heterogeneity and the uncurated nature of its content.

Unlike many datasets previously used in this field, which are typically collected either exclusively from movies or from uncontrolled real-world scenes, XD-Violence combines both types of sources. The videos come from both cinematic productions and platforms such as YouTube, including real-world scenes captured under uncontrolled conditions. The videos include indoor and outdoor environments, varying lighting conditions, multiple viewpoints, variable camera movements, and a wide range of complex visual situations. These characteristics allow the dataset to cover a broad spectrum of visual contexts and interaction dynamics, making it more representative of real-world application scenarios.

In terms of scale, XD-Violence is one of the largest datasets available in the VAD domain, surpassing even well-established benchmarks such as UCF-Crime. The large scale of the dataset is a key factor for training and evaluating models capable of effective generalization, reducing the risk of overfitting to specific contexts. [14]

Another advantage is that the dataset includes audio-visual signals, allowing the development and evaluation of multimodal approaches.

The use of the XD-Violence dataset enables the proposed system to be positioned within a framework for direct comparison with state-of-the-art approaches that employ the same benchmark and similar evaluation metrics. This allows for an objective assessment of the model's strengths and limitations, not only in terms of accuracy but also with respect to the complexity of the operational context in which it is expected to perform.

4.4 Presentation and Analysis of the Results

This section presents and analyzes the experimental results obtained from the proposed fight detection system, in accordance with the experimental protocol described in Section 4.2.

Below, the monitoring of the model's performance on the validation set is shown.

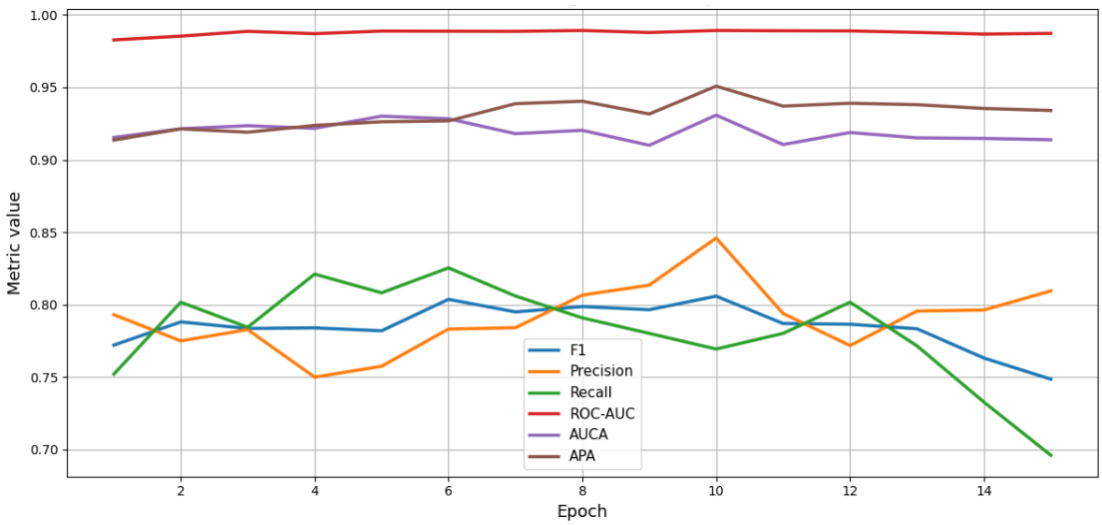

Figure 18. Evolution of training metrics over epochs

As already mentioned, particular attention is given to the AP_A metric, as it focuses exclusively on positive samples and is therefore more representative of the model's ability to detect rare events in a highly imbalanced context such as the dataset considered.

As shown in the reported graph, the model exhibits a progressive and consistent improvement in performance during the initial fine-tuning phase. Up to epoch 10, all monitored metrics display a generally increasing trend, indicating a steady refinement of the learned representations and decision boundaries.

In particular, the AP_A metric reaches its maximum value of 95.08% at epoch 10, corresponding to the model configuration that achieves the best capability to localize violent episodes within anomalous videos.

In subsequent epochs, no significant further increases in the main metrics are observed; on the contrary, some metrics, such as the F1-score, show slight fluctuations around the values previously achieved. This behavior suggests a stabilization phase in performance, during which the model has already acquired the most relevant information from the available data, and further fine-tuning iterations do not yield appreciable benefits.

In this context, the application of the early stopping strategy allowed the selection of the model corresponding to epoch 10 for the subsequent inference phase, avoiding unnecessary prolonged training and reducing the risk of overfitting to the validation data.

Inference performed on the official test set of the XD-Violence dataset, using the model selected at epoch 10, achieved overall very high performance:

AP	AP_A	AUC	AUC_A	Precision	Recall	F1 score
0.884	0.964	0.984	0.947	0.869	0.752	0.806

Table 1. Inference results

As shown in Table 1, the model achieves a ROC-AUC of 98.36% and an AP of 88.44%. The AP_A metric reaches 96.36%, confirming the model's strong ability to localize violent episodes within anomalous videos.

Precision, Recall, and F1-score values indicate a good balance between minimizing false positives and capturing most violent events.

To assess the system's applicability in real-time scenarios, the average processing times were measured at the clip level. The following results were obtained for a clip of 64 frames:

- **Data pre-processing**, which includes decoding and preparing the clip for the model, required on average approximately 340 ms.
- **Inference**, corresponding to the time taken by the V-JEPA2 model to evaluate a pre-processed clip and produce a fight/non-fight prediction, took about 937 ms.
- The **total latency**, defined as the sum of pre-processing and inference times and representing the complete time needed to decode, preprocess, analyze and classify a clip, was approximately 1,277 ms.

These timings indicate that the system is compatible with continuous monitoring scenarios using sliding temporal windows.

In terms of throughput, the system can process approximately 0.78 clips per second, corresponding to around 50 frames per second. These results indicate that the model is potentially suitable for video surveillance and continuous monitoring applications, where timely detection represents a critical requirement.

4.5 Comparison with State-of-the-Art

The objective of this section is to benchmark the proposed fight detection system against existing state-of-the-art approaches for Video Anomaly Detection. This comparison aims to contextualize the performance of the

model within the current literature and to highlight the improvements brought by the task-specific adaptations introduced in this work.

As already extensively highlighted in this thesis, most existing works in literature do not distinguish between specific types of violent events, treating all anomalies uniformly. In such approaches, the goal is typically limited to distinguishing between anomalous and normal videos, without further specialization regarding the type of event.

To ensure a fair comparison between these models and the proposed system, the evaluation procedure was adapted to focus exclusively on the detection of fight episodes. In particular, during inference on the state-of-the-art models, all other categories of anomalies present in the reference datasets (e.g., traffic accidents, explosions, shootings) were treated as normal events, so that the evaluation is consistent with the fight detection task.

For this purpose, the pre-trained weights provided by the authors of the state-of-the-art models were used, applying these ones directly to the XD-Violence test set according to the previously described protocol. The performance metrics were then recalculated based on the new anomaly definition, enabling a homogeneous and fair comparison.

In addition to comparing the proposed system with state-of-the-art VAD methods, the evaluation also included intermediate configurations specifically introduced in this work, in which the V-JEPA2 model was used solely as a video representation extractor. In these setups, the spatiotemporal features produced by V-JEPA2 were fed into simple, shallow supervised classifiers.

These classifiers have a reduced depth and a limited number of parameters compared to traditional VAD models, enabling the observed performance to be directly attributed to the quality of the video representations learned by V-JEPA2, rather than to the complexity of the downstream classification architecture.

The first classifier employed (named **CNN-1D**) is a 1D convolutional network applied along the temporal dimension of the features extracted by V-JEPA2. The spatiotemporal representations of each clip are processed through a sequence of 1D convolutions with progressively increasing stride, designed to capture local temporal patterns while gradually reducing temporal resolution.

At the end of the convolutional phase, a combined global pooling strategy integrating Global Max Pooling and Global Average Pooling is applied. This approach allows the preservation of both the most salient activations associated with short, discriminative events and the average information over the entire clip, providing a more comprehensive representation of the temporal content.

The aggregated features are finally fed to a small fully connected classifier, responsible for estimating the probability of a fighting episode occurring within the clip.

The second classifier (**CNN-1D_Attention**) extends the previous architecture by introducing a temporal attention mechanism, aiming to explicitly model the differential contribution of the various timesteps within the clip.

After feature extraction through 1D convolutional blocks, the resulting temporal sequence is processed by an attention module that assigns a

weight to each timestep. These weights, normalized via a softmax function, allow the model to focus on the most relevant temporal segments for classification, while reducing the influence of uninformative or redundant portions. The final clip representation is obtained as a weighted sum of the temporal features and is subsequently fed to a fully connected classifier to estimate the presence of a fight episode.

The third classifier tested (named **Temporal_Transformer**) is based on a temporal Transformer architecture, designed to explicitly model long-range temporal dependencies within the sequence of features extracted by V-JEPA2. In this configuration, the representations of each clip are initially projected into a lower-dimensional latent space and enriched with sinusoidal positional encodings to preserve the temporal order of frames. The sequence is then processed by a Transformer encoder composed of a limited number of layers, which captures global temporal relationships through self-attention mechanisms.

At the end of the encoder, a temporal attention pooling strategy is applied, which automatically learns the relative importance of different timesteps and produces an aggregated representation of the clip. This representation is finally used by a fully connected classifier to estimate the probability of a fighting episode occurring.

Overall, the three classifiers - a 1D CNN, a 1D CNN with temporal attention, and a temporal Transformer - provide a systematic framework to evaluate the effectiveness of V-JEPA2 representations under progressively more complex temporal modeling scenarios. All architectures operate exclusively on features pre-extracted by the backbone and do not directly process the raw video signal. Consequently, the observed performance primarily reflects the quality of the V-JEPA2

embeddings, allowing their contribution to be assessed even with relatively simple classifiers.

The table below presents a quantitative comparison between the proposed system, the three V-JEPA2-based classifiers described above, and several state-of-the-art VAD approaches, all evaluated on the specific task of fight detection.

Model	AP	AP_A	AUC	AUC_A	Precision	Recall	F1 score
V-JEPA2	**0.884**	**0.964**	**0.984**	**0.947**	**0.869**	**0.752**	**0.806**
BN-WVAD	0.852	0.962	0.988	0.813	0.833	0.751	0.79
HL-Net	0.891	0.96	0.99	0.824	0.866	0.842	0.854
UR-DMU	0.879	0.939	0.988	0.66	0.77	0.845	0.806
CNN-1D_Attention	0.819	0.918	0.969	0.911	0.823	0.671	0.739
CNN-1D	0.822	0.907	0.978	0.906	0.791	0.671	0.726
Temporal_Transformer	0.805	0.898	0.971	0.902	0.777	0.689	0.73

Table 2. Comparisons with state-of-the-art methods in XD-Violence. Results are reported in descending order of APA. V-JEPA2 denotes the fine-tuned version of the model.

The fight detection system proposed in this thesis, based on fine-tuned V-JEPA2, achieves the highest AP_A value (96.36%) and an AUC_A of 94.72%, outperforming all the state-of-the-art models considered.

These results demonstrate the system's superior ability to accurately localize fight episodes within complex video sequences.

In comparison, models such as HL-Net and BNWVAD show high global AUC values but lower performance on abnormal-specific metrics, indicating that the proposed system is more robust in handling rare and

highly imbalanced events. This confirms the effectiveness of supervised fine-tuning tailored specifically for the fight detection task.

Regarding Precision, Recall, and F1-score, the proposed model achieves high Precision (86.85%), while Recall is slightly lower than some of the competing models. This indicates a conservative strategy in labeling clips as violent, reducing false positives at the expense of a small decrease in Recall. In practical video surveillance scenarios, such a trade-off is often desirable, as it limits false alarms and increases operational reliability.

The results obtained with the simpler classifiers applied to features extracted by V-JEPA2 further highlight the quality of the backbone representations. Despite their reduced architectural complexity, these models achieve AP_A values between 89% and 92% and AUC_A values above 90%, demonstrating that V-JEPA2 features are already highly informative for the fight detection task.

However, all the three classifiers' F1-score values are lower compared to the fine-tuned V-JEPA2 model and other more complex VAD architectures. This is due to the simplicity of the classifiers, which limits their capacity to model complex temporal interactions, and consequently affects the balance between Precision and Recall at the clip-level.

5. Conclusions and future works

The experimental results presented in Chapter 4 demonstrate that the use of V-JEPA2, appropriately adapted through supervised fine-tuning, led to significant improvements in the specific task of fight detection compared to state-of-the-art VAD approaches.

A key factor contributing to this performance was the manual frame-level annotation of the training set of the XD-Violence dataset, which enabled fully supervised learning. These detailed annotations provided precise temporal supervision, allowing the model to learn discriminative patterns of violent events at clip-level.

The resulting improvement in AP_A and AUC_A metrics is particularly meaningful in the highly imbalanced context of XD-Violence, where fight episodes are rare relative to the total number of video sequences.

The evaluation protocol adopted in this work, designed to allow fair comparison with state-of-the-art models developed for VAD, further supported the significance of the results. In this context, abnormal metrics AP_A and AUC_A more accurately reflected the model's ability to localize fight episodes, whereas global AUC values can be misleading due to the predominance of normal sequences.

The comparison between the fine-tuned V-JEPA2 model and the simpler classifiers applied to the features extracted by the backbone highlighted an important trade-off between architectural complexity and clip-level classification accuracy. Although the lightweight classifiers achieved slightly lower F1-score values, they maintained high AP_A and AUC_A scores, demonstrating that the learned representations alone are highly informative for the fight detection task. The reduction in F1-score

primarily reflects the limited ability of simple classifiers to model complex temporal interactions, rather than a lack of discriminative information in the extracted features.

Computational constraints influenced several architectural choices in this work. In the case of the fine-tuned V-JEPA2 model, these constraints limited the number of backbone layers that could be effectively fine-tuned, leading to a partial unfreezing strategy in which only the final self-attention layer and the classification head were trained. When V-JEPA2 was used solely as a feature extractor, constraints limited the complexity of the downstream classifiers. However, these design choices allowed the experimentation to be completed within a reasonable timeframe while still enabling a reliable assessment of the quality of the representations learned by the backbone.

With greater computational resources, it would be feasible to unfreeze additional backbone layers and to employ deeper or more sophisticated temporal classifiers. Such enhancements could further improve clip-level classification performance while preserving the model's strong capability to localize violent events.

Looking ahead, several promising directions for future research can be identified:

1. **Extended fine-tuning of the backbone:** Unfreezing additional layers of V-JEPA2 could allow the model to adapt more effectively to fight detection, potentially improving clip-level classification while preserving high localization performance.
2. **Deeper temporal classifiers:** More sophisticated temporal modeling architectures could enhance the system's ability to

capture long-range interactions and complex dynamics within video clips.

3. **Integration of the audio information:** Although the current work focuses exclusively on the visual modality, the audio channel available in XD-Violence represents a promising complementary source of information. Incorporating audio would require developing strategies to process multimodal inputs alongside the visual stream, potentially enabling the model to better distinguish visually similar violent events, improving robustness under low-visibility conditions, and enhancing overall reliability.

4. **Extension to other anomaly types**. Since the manual frame-level annotation of the training set is complete and includes information on all anomalous events (not only fights), the pipeline designed in this work could be employed to specialize V-JEPA2 for detecting anomalies other than fights, such as shootings, protests, or car accidents. In this case, the only required modification would be in the labeling step, selecting the anomaly type of interest for clip-level supervision, while the rest of the pipeline could remain unchanged.

It is important to highlight that the manual frame-level annotation carried out on the training set of the XD-Violence dataset constitutes a key contribution of this thesis. By providing precise temporal labels for all types of anomalous events - not just fights - this fully annotated dataset represents a valuable resource for future research and addresses a common limitation in VAD studies: the lack of comprehensive, frame-level training data.

Overall, the experimental analysis confirmed that V-JEPA2, combined with manual frame-level annotations and supervised fine-tuning, constitutes a highly effective backbone for the fight detection task.

The results provided a solid foundation for developing more accurate, robust, and real-time capable systems, balancing detection performance with operational feasibility in real-world surveillance scenarios.

Acknowledgments

With deep gratitude, I would like to thank my advisors, Prof. Mario Vento and Prof. Antonio Greco, for their invaluable support, guidance, and constant availability throughout the preparation of this thesis. Their contribution has been essential to the successful completion of this work.

A heartfelt thanks goes to my family, who have always supported me with love, patience, and encouragement, making this important milestone possible.

Finally, I wish to express my gratitude to all the people I met during my university years: colleagues, friends, and professors. The experiences we shared and the relationships we built have helped me grow both academically and personally, shaping the person I am today.

List of Figures

Figure 1. Video Anomaly Detection Use Cases 8
Figure 2. VAD formulated as a weakly supervised problem using MIL. 13
Figure 3. Self/semi-supervised VAD achieved through reconstruction or prediction. 15
Figure 4. Unsupervised anomaly detection through clustering. 16
Figure 5. Illustration of abnormal frames and respective multi-modal saliencies in complex real-world scenes. 21
Figure 6. Poly-modal Induced VAD framework. 22
Figure 7. Overview of proposed model. 29
Figure 8. Original annotation structure used in the test set of XD-Violence dataset 32
Figure 9. Updated test set annotations file of XD-Violence dataset 33
Figure 10. Overview of the Vision Transformer (ViT) 36
Figure 11. V-JEPA2 architecture overview 39
Figure 12. Samples from the Something-Something dataset 41
Figure 13. Sample Classes 41
Figure 14. Sample videos from the XD-Violence dataset 44
Figure 15. Area Under the ROC Curve. 50
Figure 16. The ROC space for a "better" and "worse" classifier. 51
Figure 17. The Precision-Recall curve 53
Figure 18. Evolution of training metrics over epochs 60

List of Tables

Table 1. Inference results 61

Table 2. Comparisons with state-of-the-art methods in XD-Violence. Results are reported in descending order of AP_A. V-JEPA2 denotes the fine-tuned version of the model. 66

References

[1] B. Ramachandra, M. J. Jones and R. R. Vatsavai, "A Survey of Single-Scene Video Anomaly Detection," *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition,* 2020.

[2] G. A. Noghre, A. D. Pazho and H. Tabkhi, "A Survey on Video Anomaly Detection via Deep Learning: Human, Vehicle, and Environment," *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition,* 2025.

[3] A. Ellmetwally, R. Eldeeb and S. Elmougy, "Deep learning based anomaly detection in real-time video," *Multimedia Tools and Applications,* vol. 84, pp. 9555-9571, 2025.

[4] W. Sultani, C. Chen and M. Shah, "Real-world anomaly detection in surveillance videos.," *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition,* pp. 6479-6488, 2018.

[5] S. Majhi, G. D'Amicantonio, A. Dantcheva, Q. Kong, L. Garattoni, G. Francesca, E. Bondarev and F. Bremond, "Just Dance with π! A Poly-modal Inductor for Weakly-supervised Video Anomaly Detection," *In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition,* pp. 24265-24274, 2025.

[6] Y. Tian, G. Pang, Y. Chen, R. Sing, J. W. Verjans and G. Carneiro, "Weakly-supervised video anomaly detection with robust temporal feature magnitude learning," *In Proceedings of IEEE/CVF International Conference on Computer Vision ,* pp. 4975-4986, 2021.

[7] Y. Chen, Z. Liu, B. Zhang, W. Fok, X. Qi and Y. C. Wu, "Mgfn: Magnitude-contrastive glance-and-focus network for weakly-supervised video anomaly detection," *In Proceedings of AAAI Conference on Artificial Intelligence,* pp. 387-395, 2023.

[8] H. Lv, Z. Yue, Q. Sun, B. Luo, Z. Cui and H. Zhang, "Unbiased multiple instance learning for weakly supervised video anomaly

detection," *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition,* pp. 8022-8031, 2023.

[9] M. Cho, M. Kim, S. Hwang, C. Park, K. Lee and S. Lee, "Look around for anomalies: Weakly-supervised anomaly detection via context-motion relational learning," *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition,* pp. 12137-12146, 2023.

[10] S. Li, F. Liu and L. Jiao, "Self-training multisequence learning with transformer for weakly supervised video anomaly detection," *In Proceedings oh the AAAI Conference on Artificial Intelligence,* pp. 1395-1403, 2022.

[11] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark and e. al., "Learning transferable visual models from natural language supervision," *In International conference on machine learning,* pp. 8748-8763, 2021.

[12] H. Rasheed, M. U. Khattak, M. Maaz, S. Khan and F. S. Khan, "Fine-tuned clip models are efficient video learners," *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition,* pp. 6545-6554, 2023.

[13] Z. Li, C. Xie and E. D. Cubuk, "Scaling (down) clip: A comprehensive analysis of data, architecture, and training strategies," *arXiv preprint arXiv:2404.08197,* 2024.

[14] P. Wu, J. Liu, Y. Shi, Y. Sun, F. Shao, Z. Wu and Z. Yang, "Not only Look, but also Listen: Learnong Multimodal Violence Detection under Weak Supervision," *In European Conference on Computer Vision,* 2020.

[15] L. Zhu, L. Wang, A. Raj, T. Gedeon and C. Chen, "Advancing Video Anomaly Detection: A Concise Review and a New Dataset," *Advances in Neural Information Processing Systems,* 2024.

[16] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Komeili, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, A. Zholus, S. Arnaud, A. Gejji, A. Martin, F. R. Hogan, D. Dugas, P. Bojanowski, V. Khalidov, P. Labatut, F. Massa and M. Szafraniec, "V-JEPA 2:

Self-Supervised Video Models Enable Understanding, Prediction and Planning," *arXiv preprint arXiv:2506.09985,* 2025.

[17] R. Goyal, S. E. Kahou, V. Michalski, J. Materzyńska, S. Westphal, H. Kim, V. Haenel, I. Fruend, P. Yianilos, M. Mueller-Freitag, F. Hoppe, C. Thurau, I. Bax and R. Memisevic, "The "something something" video database for learning and evaluating visual common sense," *In Proceedings of IEEE Conference on Computer Vision and Pattern Recognition,* 2017.

[18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser and I. Polosukhin, "Attention Is All You Need," *31st Conference on Neural Information Processing Systems (NIPS 2017),* 2017.

[19] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit and N. Houlsby, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," *International Conference on Learning Representations,* 2021.

[20] L. Yuan, Y. Chen, T. Wang, W. Yu, Y. Shi, Z. Jiang, F. E. Tay, J. Feng and S. Yan, "Tokens-to-Token ViT: Training Vision Transformers from Scratch on ImageNet," *International Conference on Computer Vision and Pattern Recognition,* 2021.

[21] [Online]. Available: https://huggingface.co/docs/transformers/index.

[22] [Online]. Available: https://github.com/dmlc/decord.

[23] [Online]. Available: https://www.h5py.org/.

[24] [Online]. Available: https://pillow.readthedocs.io/en/stable/.

[25] [Online]. Available: https://en.wikipedia.org/wiki/Receiver_operating_characteristic.

[26] [Online]. Available: https://medium.com/data-science/what-is-average-precision-in-object-detection-localization-algorithms-and-how-to-calculate-it-3f330efe697b.

[27] [Online]. Available: https://medium.com/@piyushkashyap045/understanding-precision-recall-and-f1-score-metrics-ea219b908093.